



Time: 10:15 AM to 11:15 AM

Date: October 14, 2025

Max mark: 20

Note:

1. Write the time and space complexity for each program. This part is optional and has no marks.

1. Attempt ALL

- (a) Write a C program that prints the following pattern using a `while` loop:
Accept n from the user. For example $n = 5$, the output should be:

```
1 2 3 4 5
 1 2 3 4
  1 2 3
   1 2
    1
```

(5)

- (b) Write a C program to

1. Reads n integers into a dynamically allocated array.
2. Perform the following on the array :
 - i. Replace each even number with 0 and odd number with 1.
 - ii. Sort the modified array in non-decreasing order.

(1)

(4)

Example :

arr = [4, 3, 2, 1]
Output: [0, 0, 1, 1]

- Replace the even numbers (4 and 2) with 0 and the odd numbers (3 and 1) with 1. Now, arr = [0, 1, 0, 1].
- After sorting arr in non-decreasing order, arr = [0, 0, 1, 1].

- (c) Write a C program to

1. Reads n as the dimension of a $n \times n$ 2D array (matrix).
2. Perform the following on the matrix :

(1)

- i. Accept the $n \times n$ matrix

(1)

- ii. Compute and print the sum of all elements in the 0^{th} row

(1)

- iii. Compute and print the multiplication of all elements in 1^{st} column

(1)

- iv. Compute and print the sum of all elements in the anti-diagonal

(1)

- v. Compute and print the sum of all elements in the matrix

(d) Write a C program to

1. Reads n as the dimension of a $n \times n$ 2D array (matrix).
2. Perform the following on the matrix :
 - i. Print the largest element in the matrix (2)
 - ii. Print the smallest element in the matrix (2)
 - iii. Print the largest element on the diagonal (1)

>>> Repetition is the key. <<<